

Lesson 1: Charge on the move

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.1.1-2

Why this lesson is here

A lamp goes out when a wire comes loose. The charges were already in the wires: the missing connection stopped the complete path.

Learning objective

Explain what moves in a closed circuit and calculate charge flow.

Core teaching point

Current is the rate of charge flow. In one series loop, current is the same at every point; components transfer energy, not charge.

Equipment and preparation

Low-voltage cell, leads, switch, lamp, ammeter; PhET Circuit Construction Kit DC - Lab; virtual-lab worksheet pp. 3-6.

Lesson sequence

Time	Phase	Teacher move and evidence
0-5	Retrieval and case	Show a broken circuit. Ask what must be complete and what the charge carriers are.
5-14	Model	Teach conventional current, symbols and $Q = It$. Model the 3.0 min conversion.
14-30	Virtual Lab 1	Pairs predict and measure at three positions. Swap roles halfway.
30-42	Independent work	Tasks 1, 2 and 4; insist on symbol quality and units.
42-50	Check	Use the four-mark question; address 'current used up'.

Workbook task guidance

Task 1: Draw a cell, switch, lamp, resistor, ammeter and voltmeter using standard circuit symbols. Add a closed loop containing a cell, switch, lamp and ammeter.

Expected: Standard symbols: cell long/short lines; switch gap; lamp circle/cross; resistor rectangle; ammeter A circle; voltmeter V circle. Ammeter in main loop, voltmeter across a component.

Task 2: A student says current is used up by a lamp. Predict readings immediately before and after it, then explain your prediction.

Expected: Equal readings at all three series positions; charges are conserved and the lamp transfers energy.

Task 3: Virtual Lab 1: use the linked PhET lab and existing worksheet pp. 3-6. Record current at three places in one loop. Change the resistor; record what changes and what stays equal.

Expected: Lab 1 should show equal current around one loop. Increasing resistance at fixed p.d. reduces current everywhere, equally.

Task 4: Calculate charge flow for 0.25 A lasting 4.0 min. Show the time conversion and unit.

Expected: $t = 240$ s; $Q = 0.25 \times 240 = 60$ C.

Worked example

A current of 0.40 A flows for 3.0 min. $t = 180$ s; $Q = It = 0.40 \times 180 = 72$ C.

AQA-style checkpoint

[4 marks] A motor draws 0.60 A for 5.0 minutes. Calculate the charge that passes it and explain whether the current after the motor is smaller than before it.

Mark guidance: $t = 300 \text{ s}$ (1); $Q = It = 0.60 \times 300$ (1) = 180 C (1). Same current in a single loop because charge is conserved (1).

Anticipated misconception

Current is not used up. Energy is transferred by the component.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.